Error bounds for the numerical certificates of "The bias interval of the elephant random walk"

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Abstract

We give the moment, truncation, and Fourier-tail bounds used to certify the numerical threshold and critical derivatives in Remark 1.4 of the paper. These values are not used in any proof of the paper. We also give an independent fourth-moment check of bimodality near memory one.

Keywords: elephant random walk; interval arithmetic; critical derivatives.
MSC2020 subject classifications: 60G50.

This note documents the scripts in the folder `certificates` of the repository of the paper. Theorem, section, and equation numbers without an S prefix refer to the paper, arXiv:2611.14576v2.

S1 The moment series

Fix $a \in (1/2, 1)$ and write $\phi(t) = \mathbb{E}e^{it\mathcal{L}_a}$. Put $\theta_j = 1$ for even j and $\theta_j = a$ for odd j, let $m_0 = m_1 = 1$ and

$$m_n = \frac{\sum_{j=1}^{n-1} \theta_j m_j m_{n-j}}{na - \theta_n}, \qquad c_n = \frac{m_n}{\Gamma(1+an)} \quad (n \geq 2).$$

Then $\mathbb{E}\mathcal{L}_a^n = n! c_n$ [1, Theorem 1.4], with the same definition of c_0, c_1. The exponential moments in [1, Corollary 2.11] give $\phi(t) = \sum_n c_n (it)^n$ for every real t. The fixed-point equation [1, Theorem 1.1 and (2.10)] and conjugate symmetry give

$$|\phi(t)| \leq \int_0^1 |\phi(tv^a)|\,|\phi(t(1-v)^a)|\,dv. \tag{S1.1}$$

S2 Bounds for the numerical enclosures

Using the notation of Sections 1 and 3 of the paper, the moment series of Section S1, and the smoothing inequality (S1.1), we locate the threshold of Theorem 3.2 and certify its critical derivatives. Write $\Phi_N(t) = \sum_{n \leq N} c_n (it)^n$ for a partial sum of the moment series.

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S2.1 Bounds at a fixed memory

Lemma S2.1 (moment bounds)**.** *Let $N \geq 3$.*

(a) Put $A = ((N+1)a-1)/N$. If $m_j \leq AB^j$ for $1 \leq j \leq N$, then $m_n \leq AB^n$ for all $n \geq 1$.

(b) Let $S \subset \mathbb{C}$ be compact, with $\frac{1}{2} < \alpha \leq \operatorname{Re} z$, $|\operatorname{Im} z| \leq \rho$, and $|z| \leq 1$ for $z \in S$. Define $m_n(z)$ by the recursion with z in place of a, and $c_n(z) = m_n(z)/\Gamma(1+nz)$. Then every m_n is analytic near S. Put $A = ((N+1)\alpha - 1)/N$. If $|m_j| \leq AB^j$ on S for $1 \leq j \leq N$, then for all $n \geq 1$ and $z \in S$

$$|m_n(z)| \leq AB^n, \qquad |c_n(z)| \leq \frac{\tilde{A}\tilde{B}^n}{\Gamma(1+n\alpha)}, \qquad \tilde{A} = Ae^{\rho^2/(2\alpha^2)}, \quad \tilde{B} = Be^{\rho^2/(2\alpha)}.$$

Proof. For (a) we use induction on $n > N$. As $\theta_j \leq 1$ and $na - \theta_n \geq na - 1 > 0$, the bound for smaller indices gives $m_n \leq (n-1)A^2B^n/(na-1)$. The ratio $(na-1)/(n-1)$ increases in n, since its derivative is $(1-a)/(n-1)^2$, and it equals A at $n = N+1$. So $m_n \leq AB^n$.

For (b), $\operatorname{Re}(nz - \theta_n) \geq n\alpha - 1 > 0$ on S, because $\alpha \leq 1$. So m_n is analytic near S, and the induction of (a) goes through with $|\theta_j| \leq 1$. To bound the gamma factor, for $x > 0$ and real y use

$$\Big|\frac{\Gamma(x)}{\Gamma(x+iy)}\Big|^2 = \prod_{k\geq 0}\Big(1 + \frac{y^2}{(x+k)^2}\Big) \leq \exp\Big(\frac{y^2}{x^2} + \frac{y^2}{x}\Big)$$

by [2, 5.8.3]. We take $x = 1 + n\operatorname{Re} z \geq 1 + n\alpha$ and $|y| \leq n\rho$. Then $y^2/x^2 \leq \rho^2/\alpha^2$ and $y^2/x \leq n\rho^2/\alpha$. Finally $\Gamma(1 + n\operatorname{Re} z) \geq \Gamma(1+n\alpha)$, as Γ increases on $[1.47, \infty)$. □

Lemma S2.2 (truncation)**.** *Let $0 < \alpha \leq 1$, and suppose that $|c_n| \leq \tilde{A}\tilde{B}^n/\Gamma(1+n\alpha)$ for $n > N$. Let $t > 0$, $0 \leq k \leq N$, and $n_0 = N+1$, and put*

$$u = \frac{\tilde{A}\tilde{B}^{n_0} n_0^k t^{n_0-k}}{\Gamma(1+\alpha n_0)}, \qquad r = \tilde{B}t\Big(1 + \frac{1}{n_0}\Big)^k \frac{\Gamma(1+\alpha n_0)}{\Gamma(1+\alpha n_0+\alpha)}.$$

If $r < 1$, then $\big|\frac{d^k}{dt^k}\sum_{n>N} c_n(it)^n\big| \leq \tau_k(t) := u/(1-r)$, and τ_k increases on $(0,t]$.

Proof. The kth derivative of $(it)^n$ has modulus at most $n^k t^{n-k}$. The ratio of consecutive terms of the majorant $\tilde{A}\tilde{B}^n n^k t^{n-k}/\Gamma(1+\alpha n)$ is $\tilde{B}t(1+1/n)^k\Gamma(1+\alpha n)/\Gamma(1+\alpha n+\alpha)$. It decreases in n, because $\log\Gamma$ is convex. Thus the tail is bounded by the geometric series with first term u and ratio r. Both u and r increase in t. □

Lemma S2.3 (frequency cells)**.** *Let ν be even, $t_c > 0$, and $|\delta| \leq h$. Then*

$$|\phi(t_c + \delta)| \leq \sum_{k<\nu} |\phi^{(k)}(t_c)|h^k/k! + c_\nu h^\nu.$$

Proof. Taylor's formula applies with $|\phi^{(\nu)}| \leq \mathbb{E}|\mathcal{L}_a|^\nu = \mathbb{E}\mathcal{L}_a^\nu = \nu!\, c_\nu$. □

Lemma S2.4 (Fourier tail)**.** *Let $0 < T_2 < T_3$ with $T_3 \geq 2^a T_2$, and let $\beta > 3$ with $a\beta > 1$. Cover $[0, T_3]$ by cells $[l_j, r_j]$ with $l_0 = 0$ and with T_2 among the end points, and let $s_j \geq \sup_{[l_j,r_j]}|\phi|$. Let C be the largest $s_j r_j^\beta$ over the cells that meet $[T_2, T_3]$, and put*

$$R = T_2^{-1/a}\sum_{r_j \leq T_2} s_j\Big(1 - \frac{1}{2}\Big(\frac{r_j}{T_2}\Big)^{1/a}\Big)^{-a\beta}\big(r_j^{1/a} - l_j^{1/a}\big) + \frac{2^{a\beta}CT_2^{-\beta}}{a\beta - 1}.$$

If $R < 1$, then $|\phi(t)| \leq Ct^{-\beta}$ for every $t \geq T_3$.

Proof. Let $b = s_j$ on the cells in $[0, T_3)$ and $b(t) = Ct^{-\beta}$ for $t \ge T_3$. On a cell that meets $[T_2, T_3]$ we have $s_j \le Cr_j^{-\beta} \le Ct^{-\beta}$. So $b(t) \le Ct^{-\beta}$ for $t \ge T_2$, and $|\phi| \le b$ on $[0, T_3]$. Let $J(t) = \int_0^1 b(tv^a)\, b(t(1-v)^a)\, dv$.

We show $J(t) \le Rb(t)$ for $t \ge T_3$. By symmetry $J(t)$ is twice the integral over $v \le \frac{1}{2}$. There $t(1-v)^a \ge 2^{-a}T_3 \ge T_2$, so $b(t(1-v)^a) \le b(t)(1-v)^{-a\beta}$. Hence $J(t)/b(t)$ is at most $2\int_0^{1/2} b(tv^a)(1-v)^{-a\beta}\, dv$. Put $v_0 = (T_2/t)^{1/a} \le \frac{1}{2}$. On $v < v_0$ we substitute $u = tv^a$, and this part becomes

$$\frac{2}{a} t^{-1/a} \int_0^{T_2} s(u)\big(1 - (u/t)^{1/a}\big)^{-a\beta} u^{1/a-1}\, du.$$

Here $t^{-1/a} \le \frac{1}{2} T_2^{-1/a}$ and $(u/t)^{1/a} \le \frac{1}{2}(u/T_2)^{1/a}$. The weight increases in u, so on each cell we take it at r_j. As $\int_l^r u^{1/a-1} du = a(r^{1/a} - l^{1/a})$, this part is at most the first term of R. On $v \ge v_0$ we use $b(tv^a) \le Ct^{-\beta} v^{-a\beta}$ and $(1-v)^{-a\beta} \le 2^{a\beta}$. This part is at most $2^{1+a\beta} T_2^{-\beta} (T_2/t)^{1/a} C/(a\beta - 1)$, which is at most the second term of R.

Let ς be the supremum of the $\tau \ge T_3$ with $|\phi| \le b$ on $[0, \tau]$, and suppose that $\varsigma < \infty$. By continuity $|\phi| \le b$ on $[0, \varsigma]$. The arguments in (S1.1) at $t = \varsigma$ lie in $[0, \varsigma]$, so $|\phi(\varsigma)| \le J(\varsigma) \le Rb(\varsigma) < b(\varsigma)$. As $|\phi|$ and b are continuous on $[T_3, \infty)$, we get $|\phi| < b$ near ς, contradicting the definition of ς. □

This is an explicit form of Step 4 in the proof of [1, Theorem 2.5], with computed constants.

Lemma S2.5 (the second derivative)**.** *If $|\phi(t)| \le Ct^{-\beta}$ for $t \ge T_3$, with $\beta > 3$, then g_a has two continuous derivatives and*

$$g_a''(0) = -\frac{1}{\pi}\Big(\sum_{\substack{n \le N \\ n \text{ even}}} (-1)^{n/2} c_n \frac{T_3^{n+3}}{n+3} + E \Big), \qquad |E| \le \frac{T_3^3}{3}\tau_0(T_3) + \frac{CT_3^{3-\beta}}{\beta - 3}.$$

Proof. As $t^2 \operatorname{Re}\phi(t)$ is integrable, Fourier inversion gives $g_a(x) = \frac{1}{2\pi}\int e^{-itx} \operatorname{Re}\phi(t)\, dt$, and dominated convergence allows two derivatives under the integral [4, Section 3.3]. So $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\phi(t)\, dt$. On $[0, T_3]$ we replace ϕ by Φ_N, whose real part is $\sum_{n \text{ even}} (-1)^{n/2} c_n t^n$, and integrate term by term. By Lemma S2.2 the error is at most $\int_0^{T_3} t^2 \tau_0(t)\, dt \le \frac{1}{3} T_3^3 \tau_0(T_3)$. The integral over $t > T_3$ is at most $CT_3^{3-\beta}/(\beta - 3)$. □

S2.2 Bounds on memory intervals

Near the threshold, the series for ϕ cancels terms of size about 10^{60}. Direct interval evaluation would need intervals in a narrower than about 10^{-65}. We instead expand in a about one exact point, so the cancellation occurs before we bound the remainder.

Lemma S2.6 (Cauchy remainder)**.** *Let h be analytic near the closed disc $|z - a_0| \le \rho$, with $|h| \le M$ there and Taylor coefficients h_i at a_0. If $0 < \vartheta < 1$ and $|a - a_0| \le \vartheta\rho$, then $|h(a) - \sum_{i \le K} h_i (a - a_0)^i| \le M\vartheta^{K+1}/(1 - \vartheta)$.*

Proof. Cauchy's estimate gives $|h_i| \le M\rho^{-i}$. Summing the geometric tail gives the bound. □

Let $I = [a_0 - w, a_0 + w]$, $0 < \vartheta < 1$, and $\rho = w/\vartheta$, and let S be the square $a_0 + [-\rho, \rho] + i[-\rho, \rho]$, which contains the disc of Lemma S2.6. Suppose that $\alpha = a_0 - \rho > \frac{1}{2}$, that $|z| \le 1$ on S, and that $\tilde{A}, \tilde{B}$ are as in Lemma S2.1(b). Let τ_k be the bound of Lemma S2.2 with these constants. It holds for every real $a \in I$. For a cell with center t_c write $\partial_t^k \Phi_N(z, t_c) = E_k(z) + iO_k(z)$, where E_k is the sum over even n and iO_k the sum over odd n. Both E_k and O_k are real for real z. Let $E_{k,i}$ and $O_{k,i}$ be their Taylor coefficients at a_0, and put $M_k = \sum_{n \le N} \sup_S |c_n|\, n(n-1)\cdots(n-k+1)\, t_c^{n-k}$.

Proposition S2.7 (enclosure on an interval)**.** *(a) For every $a \in I$ and every t in the cell with center t_c and half-width h,*

$$|\phi_a(t)| \le \sum_{k<12} \Big(\sum_{i\le K} w^i \sqrt{E_{k,i}^2 + O_{k,i}^2} + \frac{M_k \vartheta^{K+1}}{1-\vartheta} + \tau_k(t_c) \Big) \frac{h^k}{k!} + h^{12} \sup_S |c_{12}|.$$

(b) Let T_2, $T_3 \ge 2^{a_0+w} T_2$, β, and the cells be as in Lemma S2.4, with s_j the bound of (a), and suppose that $R < 1$ for every $a \in I$. Put

$$F(z) = \sum_{n\le N,\, n \text{ even}} (-1)^{n/2} c_n(z) T_3^{n+3}/(n+3).$$

Let F_i be its Taylor coefficients at a_0, and let M_F be the same sum with $\sup_S|c_n|$ in place of $(-1)^{n/2}c_n(z)$. Then for every $a \in I$

$$\Big| g_a''(0) + \frac{1}{\pi} \sum_{i\le K} F_i (a-a_0)^i \Big| \le \eta = \frac{1}{\pi} \Big(\frac{M_F \vartheta^{K+1}}{1-\vartheta} + \frac{T_3^3}{3} \tau_0(T_3) + \frac{C T_3^{3-\beta}}{\beta - 3} \Big),$$

and $a \mapsto g_a''(0)$ is continuous on I.

Proof. For (a) we apply Lemma S2.3 with $\nu = 12$, and Lemma S2.2 to replace $\phi_a^{(k)}$ by $\partial_t^k \Phi_N$. For real a we have $|\partial_t^k \Phi_N(a, t_c)| = |(E_k(a), O_k(a))|$. Lemma S2.6 for E_k and O_k, with $\sup|E_k| + \sup|O_k| \le M_k$ on S, and the triangle inequality in $\mathbb{R}^2$ give the bracketed bound. Also $c_{12}(a) \le \sup_S |c_{12}|$.

For (b), $T_3 \ge 2^a T_2$ for every $a \in I$, so Lemma S2.4 applies at each a. Then Lemma S2.5 and Lemma S2.6 for F give the enclosure. By Lemma S2.1(b) the series for $\phi_a(t)$ converges uniformly for $a \in I$ and t in compact sets. So $\phi_a(t)$ is continuous in a. Moreover $t^2|\phi_a(t)|$ is at most t^2 on $[0, T_3]$ and $Ct^{2-\beta}$ beyond, uniformly in $a \in I$. Dominated convergence gives the continuity. □

S2.3 Critical derivative bounds

To compute the memory derivative and critical constants, we differentiate the smoothing equation and bound the resulting tail. We cannot differentiate the error bound in Lemma S2.5 to bound the derivative of the error.

Lemma S2.8 (differentiated smoothing bound)**.** *Fix a compact memory interval $I \subset (1/2, 1)$. Suppose that the cell certificate of Lemma S2.4 gives, uniformly on I, $f(t) := |\phi_a(t)| \le Ct^{-\beta}$ for $t \ge T_2$, and*

$$H_\beta(t) := 2 \int_0^{1/2} f(tv^a)(1-v)^{-a\beta}\, dv \le R < 1 \quad (t \ge T), \qquad T \ge 2^a T_2.$$

Let $\beta > 4$, $\gamma = \beta - 1$, $a\gamma > 1$, and $u = |\partial_t \phi_a|$, $w = |\partial_a \phi_a|$. Suppose that $u \le M$ everywhere, $w \le M_a$ on $[0, T_2]$, and cells give $u(t) < D_1 t^{-\beta}$ and $w(t) < D_a t^{-\gamma}$ on $[T_2, T]$. Put

$$A_1 = \frac{MC 2^{a(\beta-1)}}{a+1}, \qquad J = 2C 2^{a\beta} T^{\gamma-\beta-1/a},$$

$$\lambda = R + \frac{J T_2^{1/a-\gamma}}{a\gamma - 1}, \qquad A_0 = J M_a T_2^{1/a},$$

$$A_S = MC 2^{a(\beta-1)} \Big\{ \frac{\log 2}{a+1} + \frac{1}{(a+1)^2} \Big\} + D_1 (\log 2) R + CR/T.$$

If $A_1 + RD_1 < D_1$ and $A_0 + A_S + \lambda D_a < D_a$, with $\lambda < 1$, then the two derivative bounds hold for every $t \ge T_2$. Consequently, in the interior of I,

$$\frac{d}{da} g_a''(0) = -\frac{1}{\pi} \int_0^\infty t^2 \operatorname{Re} \partial_a \phi_a(t)\, dt, \qquad \Big| \frac{1}{\pi} \int_T^\infty t^2 \operatorname{Re} \partial_a \phi_a(t)\, dt \Big| \le \frac{D_a T^{3-\gamma}}{\pi(\gamma - 3)}. \tag{S2.1}$$

Proof. The complex majorant in Lemma S2.1(b) makes the moment series locally uniformly convergent in the frequency and memory. Cauchy's estimate on a smaller parameter disc gives the same conclusion for its memory derivative and every fixed frequency derivative. We may therefore differentiate the series for ϕ_a term by term. The smoothing equation, with $p = (1+a)/2$, is

$$\phi_a(t) = \int_0^1 \phi_a(tv^a)\{p\phi_a(t(1-v)^a) + (1-p)\phi_a(-t(1-v)^a)\}\,dv.$$

On compact frequency sets, its differentiation under the integral is justified by local boundedness of these derivatives and boundedness of $v^{\inf I}|\log v|$. Conjugate symmetry gives even moduli. Writing $x = tv^a$ and $y = t(1-v)^a$, we obtain

$$\begin{aligned}
u(t) &\le \int_0^1 \{v^a u(x)f(y) + (1-v)^a f(x)u(y)\}\,dv,\\
w(t) &\le \int_0^1 \{w(x)f(y) + f(x)w(y)\}\,dv + S(t),\\
S(t) &\le \int_0^1 \{x|\log v|u(x)f(y) + y|\log(1-v)|f(x)u(y) + f(x)f(y)\}\,dv.
\end{aligned}$$

The final term includes the derivative of p. Its coefficient is one because $p' = 1/2$ and $|\phi_a(y) - \phi_a(-y)| \le 2f(y)$.

Split each integral at $1/2$ and reflect its second half. For the frequency derivative, the smaller-argument term is at most $A_1 t^{-\beta}$, by $u \le M$ and

$$2\int_0^{1/2} v^a(1-v)^{-a\beta}\,dv \le \frac{2^{a(\beta-1)}}{a+1}.$$

The larger-argument term is at most $RD_1 t^{-\beta}$. For the memory derivative, split the smaller-argument term again at $v_0 = (T_2/t)^{1/a}$. Its bound is

$$2C2^{a\beta}t^{-\beta-1/a}\left\{M_a T_2^{1/a} + \frac{D_a T_2^{1/a-\gamma}}{a\gamma - 1}\right\}.$$

The larger-argument term is at most $RD_a t^{-\gamma}$, since $H_\gamma \le H_\beta$. As $\gamma - \beta - 1/a < 0$ and $t \ge T$, these two terms are bounded by $(A_0 + \lambda D_a)t^{-\gamma}$. In S, the smaller logarithmic term uses

$$\int_0^{1/2} v^a(-\log v)\,dv = 2^{-a-1}\left\{\frac{\log 2}{a+1} + \frac{1}{(a+1)^2}\right\};$$

the larger uses $-\log(1-v) \le \log 2$ and $H_\beta \le R$. The mixing-weight term is at most $CRt^{-\beta}$. Thus $S(t) \le A_S t^{-\gamma}$.

To extend the bounds beyond the cells, maximize the continuous weighted derivative on any $[T_2, U]$. If its maximum lies in $[T_2, T]$, use the cell bound. Otherwise repeat the preceding inequalities with that maximum in place of the proposed constant, and rearrange the strict contraction inequality. This first proves the frequency bound, then the memory bound, independently of U. The integrable uniform bound $t^2 w(t) \le D_a t^{2-\gamma}$ now permits differentiation of the Fourier formula for $g_a''(0)$ on the interior of I and gives (S2.1). □

The computation uses

$$I = [0.66881285, 0.66881287], \quad T_2 = 6, \quad T = 10, \quad \beta = 8, \quad \gamma = 7,$$

and the conservative constants

$$C = 17/1000, \quad R = 83/1000, \quad M = 2, \quad D_1 = 2, \quad M_a = D_a = 4.$$

It checks $A_1 < 1$, $\lambda < 1/10$, and $A_0 + A_S < 2$, so both contractions have strict slack. The resulting tail bound is $1/(10000\pi) < 1/30000$. The certified zero bracket lies strictly inside I.

For the finite-frequency integral, let w_0 be the half-width of I, $\rho > w_0$ the Cauchy radius, and $\vartheta = w_0/\rho$. Differentiating the finite Taylor polynomial in memory leaves the coefficient remainder factor

$$\frac{1}{\rho}\sum_{i>K} i\vartheta^{i-1} = \frac{\vartheta^K\{(K+1) - K\vartheta\}}{\rho(1-\vartheta)^2}.$$

The omitted moment series has derivative bound $\tau_k(t)/(\rho - w_0)$ by Cauchy's estimate. On each frequency cell we use a Taylor expansion of order 48, with its remainder bounded by the complex coefficient majorant. (At complex memory we cannot use a probability moment.) The signed interval checks give

$$0.68 < \frac{d}{da} g_a''(0) < 0.70, \qquad -0.09 < g_a^{(4)}(0) < -0.07 \quad (a \in \operatorname{int} I). \tag{S2.2}$$

The second enclosure uses the same Fourier bound with weight t^4, whose tail is $CT^{-3}/(3\pi)$. The file `certify_transition.py` performs these checks, including the initial derivative-cell bounds. The recorded run uses 1100 moments and 1600-bit precision under `python -O`. Each check uses `require`, so it also runs with optimization enabled.

At the unique zero from Theorem 3.2, evenness and the second inequality in (S2.2) give

$$g'_{a_*}(x) = \tfrac{1}{6} g^{(4)}_{a_*}(0) x^3 + o(x^3) < 0 \quad \text{for all sufficiently small } x > 0.$$

This agrees with Lemma B.2, which proves $g^{(4)}_{a_*}(0) < 0$ analytically. Theorem 3.2 already gives the unimodal critical shape and excludes other curvature zeros on $(1/2, 1)$.

The critical constants. For the constants C_*, B_*, and H_* of Section 4 we apply the same estimates on $I_* = [z_- - 10^{-14}, z_- + 10^{-14}]$, where

$$z_- = \frac{1838418790980431}{2748779069440000}$$

is the lower endpoint of the certified threshold bracket. The entire bracket lies inside I_*. Take

$$T_2 = 10, \quad T = 16, \quad \beta = 12, \quad \gamma = 11,$$
$$C = 1/1000, \quad R = 1/20, \quad M = D_1 = 2, \quad M_a = D_a = 4.$$

The base Fourier bound, initial derivative cells, and both strict contraction inequalities hold uniformly on I_*. We use 2000 moments, 2600-bit ball arithmetic, Cauchy ratio $1/10000$, frequency half-width $1/32$, and frequency Taylor order 48. The omitted tails for C_*, B_*, and H_* are bounded respectively by

$$\frac{D_a T^{3-\gamma}}{\pi(\gamma - 3)}, \qquad \frac{CT^{5-\beta}}{\pi(\beta - 5)}, \qquad \frac{CT^{2-\beta}}{\pi(\beta - 2)}.$$

The finite-frequency integrals use the same parameter and moment remainders as above, with weights t^2, t^4, and t. The bounds hold throughout the interval, including at a_*, whose exact value we do not need. Outward rounding gives the two derived coefficients K_* and Q_*. The record is `certificates/data/critical_constants_certificate.json`, and the README of the repository gives the command to rerun it.

S3 A fourth-moment check near one

This independent bound gives a short check of Theorem 1.2. It is not used in the proofs of the paper.

Theorem S3.1 (two modes for memory near one)**.** *If $\frac{4}{5} \le a < 1$, that is $0 < \varepsilon \le \frac{1}{10}$, then case (b) of Theorem 3.1 holds. So g_a has exactly two modes.*

Proof. In case (a), Khinchine's representation [3] gives $\mathcal{L}_a^\circ \overset{d}{=} UZ$, with U uniform on $(0,1)$ and independent of Z. Hence Cauchy–Schwarz gives $\mathbb{E}(\mathcal{L}_a^\circ)^4 \ge \frac{9}{5}[\mathbb{E}(\mathcal{L}_a^\circ)^2]^2$. The moment recursion of [1, Theorem 1.4] gives the actual ratio

$$\frac{\mathbb{E}\mathcal{L}_a^4}{(\mathbb{E}\mathcal{L}_a^2)^2} = \frac{6(2a^2+2a-1)}{4a-1}\,\mathrm{Beta}(2a,2a),$$

where Beta is the beta function. For $a \ge \frac{3}{4}$, its integral and $t(1-t) \le \frac{1}{4}$ give $\mathrm{Beta}(2a,2a) \le 4^{-2a+3/2}\,\mathrm{Beta}(\frac{3}{2},\frac{3}{2}) = \pi 2^{-4a}$. Put $R(a) = (2a^2+2a-1)/(4a-1)$. On $[\frac{3}{4},1]$ we have $(\log R)' \le 2 < 4\log 2$. The first inequality reduces to $4a(4a^2+a-2) \ge 0$. Thus $6\pi R(a)2^{-4a}$ decreases there, and for $a \ge \frac{4}{5}$ the displayed moment ratio is at most

$$\frac{282\pi}{55\,2^{16/5}} < \frac{141}{80} < \frac{9}{5},$$

using $\pi < 22/7$ and $2^{1/5} > 8/7$. Even moments are unchanged by symmetrization, so this contradicts case (a). Theorem 3.1 then gives case (b). □

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